

Intraday Options Trading

Weekly Strategy & Expiry Playbook

NIFTY (Tuesday expiry) | SENSEX / BANKEX (Thursday expiry) | NSE / BSE

1. Expiry structure and index correlation

India's two primary index option chains — NIFTY 50 and SENSEX 30 — expire on different days, creating a staggered expiry cycle each week. Unlike global markets where major indices share a common expiry, Indian weekly expiries are offset by two days:

Index	Expiry day	Constituent stocks	Key characteristic
NIFTY 50	Tuesday	50 large-caps; top 10 = ~45% weight	Higher diversification; ~50 lot size
SENSEX / BANKEX	Thursday	30 large-caps; top 10 = ~60% weight	More concentrated; higher per-stock sensitivity

1.1 The overlap problem — why they are not independent

NIFTY 50 and SENSEX 30 share approximately 25-27 large-cap constituents, including Reliance, HDFC Bank, Infosys, TCS, and ICICI Bank. These companies carry high index weight in both benchmarks. When expiry-day gamma and pin pressure forces market makers to hedge one index, they are necessarily buying or selling the same underlying stocks — which moves the other index simultaneously.

The effect is asymmetric:

- NIFTY expiry (Tuesday): SENSEX absorbs 60-70% of the pin/volatility because the shared large-caps dominate SENSEX's more concentrated 30-stock composition.
- SENSEX expiry (Thursday): NIFTY absorbs 40-50% because its 50-stock breadth provides more diversification and dampens the transmission.

1.2 Practical implications for strategy selection

On any expiry day, the non-expiring index is not a clean, independent alternative. It receives synthetic pin pressure from the expiring index's gamma unwind. This creates a two-phase pattern every expiry day:

Phase	Non-expiring index behaviour	Recommended strategies
9:15 – 11:30 AM	Trades cleanly. Full premium. ORB levels are valid.	ATM_ORB, ATM_MOMENTUM freely

11:30 AM – 2:00 PM	Gamma interference zone. Chop + mean-reversion patterns. Stops get hit unpredictably.	ATM_TRAP_REVERSAL, ATM_VWAP_REVERSION only
2:00 – 3:30 PM	Pin and reversal dynamics peak. Wide bid-ask. Non-expiring index still affected.	ATM_TRAP_REVERSAL only on non-expiring index

1.3 Post-expiry IV reset — a trading opportunity

When an index expires (NIFTY on Tuesday, SENSEX on Thursday), its ATM implied volatility collapses as premium evaporates. Because the indices share underlying stocks, the non-expiring index also sees a mild IV compression. The day after expiry, when the non-expiring index starts fresh, its IV resets to a cleaner baseline.

This creates the single best ATM_IV_EXPANSION setup of the week:

- Wednesday morning (day after NIFTY Tuesday expiry): NIFTY next-week ATM IV has reset cleanly. First expansion body candle of the session fires with high reliability. This is a priority ATM_IV_EXPANSION entry.
- Friday morning (day after SENSEX Thursday expiry): SENSEX next-week ATM IV reset. Same pattern — ATM_IV_EXPANSION on SENSEX Friday morning is a priority setup.

2. Day-by-day strategy plan

The table below maps each weekday to the correct primary engine, recommended strategies, and restrictions. All strategy names correspond exactly to the intraday_options engine's supported strategies in the bot's configuration.

Mond ay No expiry today NIFTY: 2 DTE (caution) SENSEX: 4 DTE (primary)	Tuesday NIFTY EXPIRY NIFTY: EXPIRY SENSEX: 3 DTE	Wednesday No expiry today NIFTY: 5 DTE (best day) SENSEX: 2 DTE (caution)	Thursday SENSEX EXPIRY NIFTY: 4 DTE SENSEX: EXPIRY	Friday No expiry today NIFTY: 3 DTE (fresh) SENSEX: 5 DTE (fresh)	
Primary engine	SENSEX is primary. NIFTY next-week only.	SENSEX only. NIFTY current-week: avoid.	NIFTY next-week is primary. SENSEX: pre-noon only.	NIFTY next-week only. SENSEX: avoid entirely.	Both indices fresh. Cleanest day of the week.
Strategies	ATM_ORB (SENSEX) ATM_MOMENTUM (SENSEX) ATM_TRAP_REVERSAL (NIFTY next) ATM_VWAP_REVERSAL (NIFTY, pre-noon)	ATM_ORB (SENSEX) ATM_MOMENTUM (SENSEX) ATM_IV_EXPANSION (SENSEX) NIFTY next-week: TRAP_REVERSAL only	ATM_ORB (NIFTY next) ATM_MOMENTUM (NIFTY next) ATM_BREAKOUT_EXPANSION (NIFTY) ATM_IV_EXPANSION (NIFTY - post-expiry reset) ATM_TRAP_REVERSAL (SENSEX, morning)	ATM_MOMENTUM (NIFTY next) ATM_ORB (NIFTY next) ATM_MULTI (NIFTY - higher confirmation for noise) Pre-noon: full strategy set on NIFTY	ATM_ORB (both) ATM_MOMENTUM (NIFTY) ATM_IV_EXPANSION (SENSEX - post-expiry reset) ATM_BREAKOUT_EXPANSION (SENSEX) ATM_MULTI (either)
Cautions	NIFTY current-week: reduce size, prefer next-week contract.	NIFTY current-week: DO NOT TRADE after 10:30 AM. Gamma explosion, pin risk, wide spreads.	SENSEX: pre-noon only, reduce size, reversals only post 11 AM.	SENSEX/BANKEX current-week: DO NOT TRADE. Gamma explosion. NIFTY post 2 PM: TRAP_REVERSAL only.	No expiry risk. Avoid new CNC delivery positions in the afternoon (weekend gap risk).

3. Detailed daily notes

Monday — NIFTY 2 DTE, SENSEX 4 DTE

Monday is a split-personality day. SENSEX has full four-day premium and trades cleanly — it is your primary engine. NIFTY is already entering its final two days before Tuesday expiry, meaning theta acceleration begins this afternoon and pin pressure starts building around key strikes.

Primary engine: SENSEX (4 DTE) — run ATM_ORB and ATM_MOMENTUM freely all session.

NIFTY strategy: Switch to next-week NIFTY contract. Only ATM_TRAP_REVERSAL and ATM_VWAP_REVERSION on NIFTY, pre-noon only.

Size rule: Full size on SENSEX. Reduced size (50%) on any NIFTY current-week position.

Config: max_trades_per_underlying=2, time_exit_minutes=20

Afternoon watch: As NIFTY current-week approaches expiry tomorrow, bid-ask spreads widen post 1 PM. Any open NIFTY current-week positions should be managed tightly or closed by 2 PM.

Tuesday — NIFTY EXPIRY, SENSEX 3 DTE

Tuesday is NIFTY expiry day. The single most important rule: do not trade NIFTY current-week options after 10:30 AM. Gamma risk is extreme, strikes act as magnets, and bid-ask spreads make profitable exits unreliable.

Primary engine: SENSEX (3 DTE) — this is a prime SENSEX trading day.

Key insight: NIFTY expiry creates directional volatility that SENSEX mirrors (60-70% correlation). This makes ATM_IV_EXPANSION on SENSEX extremely effective — the expiry-driven body candles on SENSEX frequently satisfy the 1.8x body threshold.

NIFTY next-week: ATM_TRAP_REVERSAL only, if at all. The expiry volatility bleeds into next-week contracts too.

Config: max_trades_per_underlying=2 (SENSEX), 0 (NIFTY current-week), time_exit_minutes=18

Hard rule: No NIFTY current-week options after 10:30 AM on Tuesday. No exceptions.

Wednesday — NIFTY next 5 DTE, SENSEX 2 DTE

Wednesday is the best NIFTY trading day of the entire week. The next-week NIFTY contract just entered its first full day — maximum premium, clean ATM levels, no pin pressure. Run the full NIFTY strategy set without restriction.

Primary engine: NIFTY next-week (5 DTE) — best conditions of the week.

ATM_IV_EXPANSION priority: Wednesday morning is the highest-reliability ATM_IV_EXPANSION setup of the week. Post-NIFTY expiry IV reset creates a clean baseline; the first expansion candle is a strong signal. Treat this as a priority entry.

ATM_BREAKOUT_EXPANSION: Also viable on NIFTY Wednesday — full 45-candle compression can form, and premium supports wide targets.

SENSEX (2 DTE): Pre-noon only. ATM_TRAP_REVERSAL in the morning session. Avoid post-noon SENSEX — theta bites hard with 2 DTE and tomorrow's expiry.

Config: max_trades_per_underlying=2 (NIFTY), 1 (SENSEX pre-noon), time_exit_minutes=18

Thursday — NIFTY next 4 DTE, SENSEX EXPIRY

Thursday mirrors Tuesday in structure but with SENSEX expiring instead of NIFTY. SENSEX/BANKEX current-week options must not be traded. NIFTY next-week (4 DTE) is your sole options engine.

Primary engine: NIFTY next-week (4 DTE) — strong premium, full strategy set available.

ATM_MULTI preferred: SENSEX expiry creates noise that bleeds into NIFTY price action (40-50% correlation). ATM_MULTI's higher confirmation filter (requires aligned ORB + momentum) protects against false signals during this noise window, particularly in the 11:30 AM – 2 PM interference zone.

Morning session (pre 11:30): Run ATM_ORB and ATM_MOMENTUM on NIFTY freely — clean directional trades before the SENSEX expiry unwind intensifies.

Afternoon (post 2 PM): ATM_TRAP_REVERSAL only on NIFTY. The expiry dynamics create pin-and-reversal setups around SENSEX strikes that also affect NIFTY price action.

Config: max_trades_per_underlying=2 (NIFTY only), time_exit_minutes=20

Hard rule: No SENSEX/BANKEX current-week options at any time on Thursday. No exceptions.

Friday — NIFTY next 3 DTE, SENSEX next 5 DTE

Friday is the cleanest trading day of the week. Both contracts are fresh — NIFTY at 3 DTE, SENSEX at its very first day of the new week's cycle. No expiry noise, no gamma interference, no pin pressure on either index. The full strategy set is available for both underlyings.

Primary engine: Both NIFTY and SENSEX — equal priority.

ATM_IV_EXPANSION on SENSEX: Friday morning is the best ATM_IV_EXPANSION setup for SENSEX. Post-Thursday expiry IV reset creates the same clean baseline as Wednesday's NIFTY setup. Treat as a priority entry.

ATM_BREAKOUT_EXPANSION on SENSEX: Also high priority — fresh contract, new weekly cycle, compression setups form cleanly from session open.

No expiry restrictions: Run full max_trades on both underlyings. Full time_exit window.

Delivery (CNC) note: Avoid new CNC entry positions Friday afternoon — weekend gap risk. Hold existing positions.

Config: max_trades_per_underlying=2 (both), time_exit_minutes=20

4. Time-of-day rules

These rules apply every day, independent of which index you are trading. They reflect the natural intraday rhythm of Indian options markets and are structured around your bot's existing entry cutoff of 15:10 (LEGACY_IMMEDIATE mode).

Session	9:15 – 10:30 ORB window	10:30 – 13:00 Trend session	13:00 – 15:10 Late session
Strategies	ATM_ORB (primary) ATM_MOMENTUM (secondary) Wait for 15 candles	ATM_MOMENTUM ATM_BREAKOUT_EXPANSION ATM_MULTI ATM_IV_EXPANSION	ATM_VWAP_REVERSION only ATM_TRAP_REVERSAL only Hard cutoff: 15:10
Config	time_exit_minutes: 20-25 max_trades: 2	time_exit_minutes: 18-20 max_trades: 2	time_exit_minutes: 10-15 max_trades: 1

5. Strategy reference — DTE suitability

The table below summarises each intraday_options strategy with its DTE suitability, entry window, and notes. Strategies that require large premium budgets are progressively less suitable as DTE shrinks.

Strategy	Type	Entry window	Core logic	Best days	Notes
ATM_ORB	Trend / Breakout	First 15 candles	CE above ORB high, PE below ORB low	Day 1-3 (Fri-Wed)	Full premium required — 3+ DTE
ATM_MOMENTUM	Trend follow	Any time (RSI+VWAP)	CE on bullish, PE on bearish	Day 1-3 (Fri-Wed)	Needs room to run — avoid 1 DTE
ATM_BREAKOUT_EXPANSION	Compression breakout	Post 10:30 AM	45-candle compression + volume + ATR	Day 2-3 (Mon-Wed)	Needs 45 candles to form; skip on 1-2 DTE
ATM_MULTI	High-confidence combo	Post 10:30 AM	ORB + momentum aligned	Day 2-3, expiry noise days	Higher filter — best when VWAP/expiry noise present
ATM_IV_EXPANSION	IV event capture	Post 10:30 AM	Body $\geq 1.8x$ avg + RSI	Tue (SENSEX), Fri (SENSEX)	Post-expiry IV reset days produce best signals
ATM_VWAP_REVERSION	Mean reversion	Any session	6-candle VWAP deviation + re-entry	Wed-Thu (short holds)	Pre-noon on 1-2 DTE days; 15-min max hold
ATM_TRAP_REVERSAL	Stop-hunt reversal	Any session	S/R trap + 3 candle + body $\geq 1.5x$	Mon, Thu (pre-expiry)	Best on pre-expiry and expiry eve days

Key principle: ATM_ORB, ATM_MOMENTUM, and ATM_BREAKOUT_EXPANSION all require the option premium to have enough range to hit a 12-20% target after stops of 8-12%. On a 1-DTE contract, the same underlying move produces only 30-40% of the premium movement versus a 4-5 DTE contract. The strategy set shifts from trend-following and expansion on fresh premium days to reversal and mean-reversion as DTE collapses.

6. Bot configuration — per-day quick reference

The following settings should be adjusted each morning before starting the session. All values are set from the bot's web UI under the configure form or session override controls.

Day	Primary engine	max_trades	Restrictions	time_exit (min)	Notes
Monday	SENSEX (4 DTE)	2	NIFTY next preferred	20	SENSEX primary. NIFTY: next-week contract only.
Tuesday	SENSEX (3 DTE)	2 SENSEX / 0 NIFTY curr.	NIFTY next-week only	18	No NIFTY current-week after 10:30 AM.
Wednesday	NIFTY next (5 DTE)	2 NIFTY / 1 SENSEX	Pre-noon SENSEX only	18	Best NIFTY day. SENSEX: pre-noon + reversals only.
Thursday	NIFTY next (4 DTE)	2 NIFTY only	No SENSEX	20	SENSEX expiry. NIFTY next-week is sole engine.
Friday	Both fresh	2 each	Full strategy set	20	Cleanest day. Both contracts fresh. No expiry noise.

6.1 Standing configuration — every day

- Options stop loss: 8-12% (never tighten below 8% — bid-ask spread consumes the buffer)
- Options target: 12-20% (minimum 12% to clear total transaction costs and slippage)
- Lot sizes: NIFTY = 75 lots, SENSEX = 20 lots (verify before each session — SEBI revises periodically)
- Entry mode: LEGACY_IMMEDIATE (confirmed working; staged modes require additional validation)
- Entry cutoff: 15:10 (hard stop — no new entries after this time regardless of signal)
- ATM roll trigger: configured pct (default) — allow automatic rolling when ATM shifts by the trigger threshold

6.2 Mid-day strategy switch rules

Mid-day strategy switching is permitted and sometimes necessary (e.g., switching from ATM_MOMENTUM to ATM_TRAP_REVERSAL post 1 PM). Follow these rules:

- Close or confirm all open positions before switching strategy — do not have simultaneous open positions from different strategies in the same underlying.
- Do not switch to a strategy whose minimum candle requirement (e.g., 45 candles for ATM_BREAKOUT_EXPANSION) has not yet been met for the day.
- When switching post-noon to ATM_VWAP_REVERSION or ATM_TRAP_REVERSAL, also reduce time_exit_minutes to 10-12 and max_trades to 1.

7. Risk and position management rules

7.1 Expiry-day position sizing

On expiry days (Tuesday for NIFTY, Thursday for SENSEX), apply the following size adjustments to the expiring index:

- Current-week options: do not open new positions. If already open at session start, apply `safe_mode` immediately — stop to breakeven, half target, trailing active.
- Any open current-week position not closed by 2 PM should be exited at market via the `emergency_exit` override in the bot's UI.

7.2 Premium viability check (cost-to-profit ratio)

Your bot already gates entries on `max_entry_cost_ratio`. On late-DTE days the natural premium shrinkage means cost-to-profit ratio deteriorates even for valid signals. A manual check before each entry on 1-2 DTE days:

- Entry price above 50: viable for 8-12% stop, 12-20% target structure.
- Entry price 20-50: viable only with 10-12% stops and 18-20% targets — tighter stops will be hit by spread alone.
- Entry price below 20: do not enter. The absolute rupee move needed to hit target is smaller than typical bid-ask spread on expiry week.

7.3 Correlation guard on simultaneous positions

Because NIFTY and SENSEX are 60-70% correlated, holding a NIFTY CE and a SENSEX CE simultaneously is effectively a doubled position in the same directional bet. On non-expiry days this is acceptable if both signals are independent. On expiry days and the pre-expiry day (Monday for NIFTY, Wednesday for SENSEX), treat simultaneous positions in both indices as a single combined exposure and apply `max_open_positions` limits accordingly.

7.4 Trailing stop and tick-exit interaction

Your bot's tick-exit monitor arms real-time WebSocket breakout callbacks for `intraday_options`. The trailing stop breakeven clamp ensures the trailing stop never activates below entry price. On expiry days, the rapid premium swings create additional value from the tick-exit trailing ratchet — the intra-candle spike capture is especially important when option premiums can move 15-20% within a single minute on expiry pin-reversal moves. Do not disable `tick_exit` for `intraday_options` on expiry days.

8. Weekly flow summary

A single-paragraph summary of the optimal weekly flow for quick reference each Monday morning:

Monday: SENSEX is your workhorse (4 DTE). Run ATM_ORB and ATM_MOMENTUM on SENSEX all day. Switch NIFTY to next-week contract and use only ATM_TRAP_REVERSAL pre-noon. NIFTY current-week expires tomorrow — reduce exposure.

Tuesday: NIFTY expires. Do not trade NIFTY current-week after 10:30 AM. Flip entirely to SENSEX (3 DTE). ATM_ORB, ATM_MOMENTUM, and ATM_IV_EXPANSION on SENSEX — expiry volatility amplifies SENSEX body candles and makes IV expansion signals fire reliably.

Wednesday: Best NIFTY day. Fresh next-week contract at 5 DTE. ATM_IV_EXPANSION Wednesday morning is the priority setup of the week — post-expiry IV reset. Run full strategy set on NIFTY. SENSEX is 2 DTE — pre-noon reversals only.

Thursday: SENSEX expires. Mirror of Tuesday — NIFTY next-week (4 DTE) is sole engine. Use ATM_MULTI to filter noise from SENSEX expiry bleed. No SENSEX trades at all. Strong morning session on NIFTY before 11:30 AM.

Friday: Cleanest day. Both contracts fresh. ATM_IV_EXPANSION on SENSEX (post-expiry reset) and ATM_BREAKOUT_EXPANSION on SENSEX are priority setups. Run full strategy set on both indices. No expiry noise anywhere this session.

9. Morning decision checklist

Run through this checklist before starting each session:

- **Step 1 — Identify the day:** Is today an expiry day? (Tuesday = NIFTY expires, Thursday = SENSEX expires)
- **Step 2 — Set primary engine:** Use the table in Section 2 to confirm which index is primary today.
- **Step 3 — Configure the bot:** Set max_trades_per_underlying, time_exit_minutes per Section 6 table.
- **Step 4 — Check VIX:** VIX > 18: reduce position size 25%. VIX > 22: reduce 50%, use ATM_MULTI/ATM_VWAP_REVERSION only. VIX < 12: reduce size 25% (low premium environment).
- **Step 5 — Morning strategy:** Set ATM_ORB + ATM_MOMENTUM for 9:15-10:30 ORB window.
- **Step 6 — Mid-day review (12:30 PM):** Switch to ATM_VWAP_REVERSION or ATM_TRAP_REVERSAL for afternoon. Reduce time_exit to 10-12 min.
- **Step 7 — Expiry guard (expiry days only):** Check all current-week positions of expiring index by 10:15 AM. Close or apply safe_mode.

This document reflects the current understanding of NIFTY-Tuesday / SENSEX-Thursday expiry structure and the index correlation dynamics described above. Review and update if expiry schedules change (SEBI periodically revises weekly expiry days).

10. Trading expiry-day options — the low-premium playbook

Some traders deliberately seek out the expiring contract on its expiry day. The appeal is straightforward: elevated gamma means that even a small underlying move produces a large percentage change in the option premium, and the short-dated contract's tight time-value decay window can amplify directional moves into 30-50% option gains in minutes. However, the same mechanics that create this opportunity also create extreme risk. This section provides a structured approach for anyone who chooses to trade expiry-day contracts deliberately.

10.1 Why premiums collapse on expiry day — the mechanics

An option premium is composed of intrinsic value (how far in-the-money it is) and time value (the probability of further movement before expiry). On expiry day, time value decays to near zero. An ATM option that was worth Rs 150 five days ago may be worth Rs 20-40 on expiry morning. This creates three specific hazards:

Hazard	What happens	Impact on bot strategy
Bid-ask spread dominance	Spread (e.g. Rs 18 bid / Rs 24 ask) is now 25-33% of the option price. Entry at ask + exit at bid = immediate 30% loss before the market moves.	Your 8-12% stop config fires on spread noise alone. Stops must be widened to 15-20% minimum.
Gamma explosion	Delta and gamma shift violently near ATM. A 10-point NIFTY move can move the ATM option 15-25% in seconds, then snap back as pin pressure reasserts.	Tick-exit trailing ratchet is critical here. Without it, a spike-and-crash candle captures nothing. Ensure tick_exit is active.
Strike pinning	Market makers pin the underlying near maximum-pain strike in the final 2 hours. The index oscillates around this level, creating repeated false breakouts.	ATM_ORB and ATM_MOMENTUM fail here — the breakout is reversed within minutes. Only reversal and reversion strategies are reliable.
Liquidity collapse	OTM strikes become near-worthless. ATM liquidity concentrates at the maximum-pain strike. Strikes more than 0.3% away have negligible volume.	Bot's OI filter and spread check must pass. Only pure ATM strikes qualify. Rolling to a new ATM is extremely costly on expiry day.

10.2 When expiry-day trading can work

Despite the hazards, expiry-day options have a legitimate place in an intraday strategy. The two scenarios where the gamma amplification works in your favour rather than against you:

Scenario	Why it works	Strategy to use
Strong directional open — index opens 0.4%+ away from prior close and sustains direction in first 15 candles	Pin pressure breaks down when the move is large enough to overcome max-pain gravity. Gamma amplifies the move — a 0.5% NIFTY move = 40-60% ATM option gain.	ATM_ORB only. Strict ORB levels from first 15 candles. CE above ORB high or PE below ORB low. Exit as soon as momentum stalls — do not hold.
Clear pin-and-reversal setup — index tests max-pain strike, fails to	Max-pain pinning is a deliberate institutional behaviour. Once a test of the pin level fails with a reversal body candle, the move back is fast and clean.	ATM_TRAP_REVERSAL exclusively. 3-candle trap confirmation required. Hold time maximum 10-12 minutes. Take partial exit at first 15-20% gain.

break, reversal candle forms within 3 candles		
Post-max-pain VWAP snap — index breaks away from VWAP significantly in first 30 min, then snaps back sharply	On expiry day, VWAP acts as an unusually strong gravity well. Deviations of 0.3%+ from VWAP in the first 30 min almost always snap back within 2-3 candles.	ATM_VWAP_REVERSION. Pre-noon only. Confirm 6-candle deviation then re-entry through VWAP. Short hold — exit at first VWAP touch.

10.3 Mandatory configuration changes for expiry-day contracts

Your standard intraday_options configuration must be overridden when trading the expiring contract. These are not optional adjustments — the default config is calibrated for 3-5 DTE contracts. Using it unchanged on expiry day will result in stops being triggered by spread noise before the market has a chance to move.

Parameter	Normal config (3-5 DTE)	Expiry-day override	Reason
Stop loss (options)	8-12%	15-20%	Bid-ask spread alone is 20-30% of premium. An 8% stop fires on spread noise before any adverse move.
Target (options)	12-20%	25-40%	Gamma amplification means a valid setup can hit 30-50% quickly. A 12% target exits too early and misses the primary move.
time_exit_minutes	15-20 min	8-12 min	Premium decays by the minute. A position that hasn't moved in 12 min will have lost 5-10% of its value to theta alone.
max_trades_per_underlying	2	1	One loss on an expiry contract can be large in absolute terms. A second trade on the same underlying compounds risk on a day with extreme variance.
Position size (capital per trade)	Full max_capital_per_trade	50% of normal	The 15-20% stop on a full position doubles your absolute loss exposure vs normal. Halve the size to keep per-trade loss equivalent.
Entry mode	LEGACY_IMMEDIATE	LEGACY_IMMEDIATE (keep)	Staged entry confirmation is too slow on expiry day — the move is often over by the time staged filters confirm. Keep immediate entry.
Lot mode	Capital-based or One-lot	ONE_LOT mandatory	Capital-based sizing on a Rs 20-40 option inflates lot count dangerously. ONE_LOT caps absolute exposure.

10.4 Allowed and forbidden strategies on expiry day

Not all seven intraday_options strategies are viable on expiry-day contracts. The table below gives an unambiguous verdict for each, with the reason rooted in the strategy's minimum candle requirements, signal logic, and premium budget assumptions.

Strategy	Verdict	Why	Condition if allowed
ATM_TRAP_REVERSAL	ALLOWED	Specifically designed for stop-hunt and pin-reversal dynamics. Expiry day maximises these setups. 3-candle trap confirmation limits false signals.	Apply expiry-day config (15-20% SL, 25-40% target, 8-12 min exit). Only pre-2 PM.
ATM_VWAP_REVERSION	ALLOWED (pre-noon)	VWAP gravity is unusually strong on expiry day. Pre-noon VWAP deviations revert cleanly. Post-noon VWAP becomes unreliable as pin pressure dominates.	Pre-noon only. Max hold 10 min. Exit at first VWAP touch — do not hold for full target.
ATM_ORB	CONDITIONAL	ORB works only if the opening 15-candle range is wide (large gap or sustained directional momentum). On a flat open, ORB levels are too tight and will be violated repeatedly by pin action.	Only if ORB range > 0.4% of index level. Discard if range is tight. Morning session only (pre 11 AM). Expiry-day config mandatory.
ATM_MOMENTUM	AVOID	RSI + VWAP trend follow requires a sustained trend. Pin pressure causes repeated trend reversals on expiry day. The strategy generates signals that appear valid but are reversed within 2-3 candles.	Do not use on expiry-day contracts. Use on the non-expiring index instead.
ATM_MULTI	CONDITIONAL	Requires both ORB and momentum to align. On expiry day, momentum component frequently fails the filter, reducing the number of valid setups to near zero. However, when ATM_MULTI does signal on expiry day, it is a high-quality setup because both conditions have overcome pin pressure.	Only if ATM_MULTI signals before 11 AM. Expiry-day config. Treat as a priority trade when it does fire.
ATM_BREAKOUT_EXPANSION	FORBIDDEN	Requires 45-candle compression and ATR expansion. On expiry day, ATR is artificially elevated by pin oscillations — the compression detection fires false signals. Premium cannot sustain the target distance after bid-ask cost.	Do not use on any expiry-day contract under any circumstances.
ATM_IV_EXPANSION	FORBIDDEN	IV is collapsing on expiry day, not expanding. This strategy requires expanding implied volatility. Entering an IV expansion strategy on a contract in IV collapse is a fundamental logic contradiction.	Do not use on expiry-day contracts. ATM_IV_EXPANSION belongs on fresh contracts (Wednesday NIFTY, Friday SENSEX) only.

10.5 Time windows — expiry-day specific

The intraday structure of an expiry day is different from a normal trading day. Pin pressure builds in phases. The time windows below replace the standard time-of-day rules from Section 4 on expiry-day contracts.

Time window	Market behaviour	Allowed strategies	Notes
9:15 – 10:30	Directional open possible. ORB forming. Pin not yet fully active.	ATM_ORB (if range > 0.4%), ATM_MULTI, ATM_VWAP_REVERSAL	Best window for expiry-day trading. Highest probability setups occur here. If no valid signal by 10:30, seriously consider skipping the expiry contract entirely.
10:30 – 13:00	Pin pressure builds. Max-pain gravity intensifies. Choppy oscillation around key strikes.	ATM_TRAP_REVERSAL, ATM_VWAP_REVERSAL (pre-noon only)	Reduce to 1 trade max in this window. Set time_exit_minutes=8. Anticipate the trap setup near max-pain strike.
13:00 – 14:30	Maximum pin intensity. Wide spreads. Violent gamma swings. Near-zero premium on losing side of strike.	ATM_TRAP_REVERSAL only	Experienced traders only. Use only if a very clear trap candle pattern forms at max-pain strike. Half normal size even with expiry config.
14:30 – 15:10	Final settlement oscillation. Extreme spread. Gamma at peak. Stops triggered randomly.	DO NOT TRADE	No new entries after 14:30 on the expiring contract. If still holding an open position, exit immediately at market regardless of P&L.

10.6 Premium viability gate — do not enter below this price

Before placing any trade on an expiry-day contract, check the current ATM option price. The minimum viable premium is calculated from your stop distance, target, and the absolute rupee amount needed to exceed total transaction costs (brokerage, STT, exchange fees, slippage).

Entry price range	Verdict	Lot mode	Notes
Below Rs 15	DO NOT ENTER	Any	Total transaction cost (brokerage + STT + exchange + slippage) exceeds any realistic 25-40% gain. The spread alone (typically Rs 4-6) is 25-40% of premium.
Rs 15 – Rs 35	CAUTION	ONE_LOT only	Borderline viable. 15-20% SL and 25-40% target still work mathematically but margin for error is minimal. ONE_LOT mandatory. Only ATM_TRAP_REVERSAL at this level.
Rs 35 – Rs 80	VIALE	ONE_LOT, 50% capital	Good premium range for expiry-day setups. Absolute gain per lot is meaningful. Spread is now < 15% of premium. Apply expiry-day config: 15-20% SL, 25-40% target.
Above Rs 80	VIALE — recheck strikes	ONE_LOT, 50% capital	Premium > Rs 80 on expiry day typically means the option is ITM, not ATM. Verify

the strike is truly at-the-money. ITM options have lower gamma sensitivity and behave more like the underlying stock — not the high-gamma setup you are looking for.

10.7 Tick-exit monitor — non-negotiable on expiry day

Your bot's tick-exit system (Section 6 of the README) arms WebSocket breakout callbacks that capture intra-candle spikes. On expiry day this feature transitions from useful to essential. The following scenario illustrates why:

Example: NIFTY ATM CE entered at Rs 42. Within one 1-minute candle, NIFTY surges 25 points — the option spikes to Rs 68 (62% gain). NIFTY then snaps back 20 points as pin pressure reasserts — the option closes the candle at Rs 31 (a loss). Without tick-exit, the bot sees only the candle close at Rs 31 and exits at a loss. With tick-exit trailing ratchet, the trailing stop ratchets from Rs 42 to approximately Rs 55 during the spike, and fires on the way down — exit at ~Rs 55, capturing a 31% gain from the same candle that would have been a 26% loss.

Verify before each session: tick_exit is enabled for intraday_options in engine_config.py. Do not trade expiry-day contracts if the WebSocket connection to Kite is unavailable — the REST LTP fallback polls every 3-10 seconds, which is too slow to capture expiry-day gamma spikes that complete within one or two candles.

Hard rule: If KiteConnect WebSocket is showing disconnected at session start and cannot reconnect within 5 minutes, do not trade expiry-day contracts. Trade the non-expiring index only where the candle-close exit cadence is sufficient.

10.8 Expiry-day trading — pre-session checklist

Before initiating any trade on the expiring contract, confirm all of the following:

- **Stop loss set to 15-20%:** Not 8-12%. Verify in bot config before market open.
- **Target set to 25-40%:** Not 12-20%. Expiry-day gamma requires wider target to capture the real move.
- **Lot mode set to ONE_LOT:** Capital-based sizing is dangerous on low-premium contracts. ONE_LOT is mandatory.
- **Position size at 50% of normal:** Reduce max_capital_per_trade to 50% before session start.
- **max_trades_per_underlying = 1:** One trade on expiry contract. If it loses, move to non-expiring index.
- **time_exit_minutes = 8-12:** Not 15-20. Theta decay is rapid. Exit non-performing positions early.
- **KiteConnect WebSocket is connected:** Tick-exit is non-negotiable. If WebSocket is down, skip expiry contract.
- **Entry price > Rs 35:** Below Rs 35, costs exceed realistic gains. Skip and wait for a better setup.
- **ATM_BREAKOUT_EXPANSION and ATM_IV_EXPANSION are disabled:** These strategies must not be used on expiry-day contracts.

- **Strategy selected matches the time window:** ATM_ORB / ATM_MULTI before 11 AM. ATM_TRAP_REVERSAL / ATM_VWAP_REVERSION 11 AM - 1 PM. No new trades after 2:30 PM.

Final caution: Expiry-day contracts are a specialist instrument within an already complex strategy. The default recommendation is to leave expiry-day contracts to the non-primary index — trade NIFTY next-week on Tuesday, SENSEX next-week on Thursday — and only deliberately trade the expiring contract if you have backtested the specific setup, the tick-exit system is confirmed active, and all configuration overrides above are applied. The profit potential is real but the loss scenarios are equally amplified.